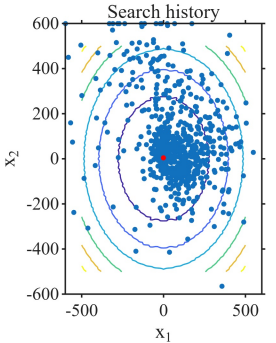
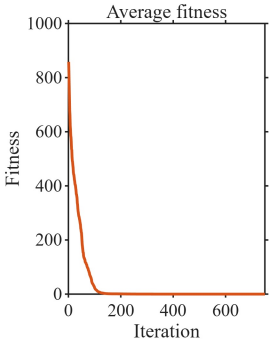
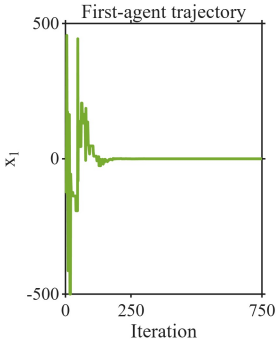
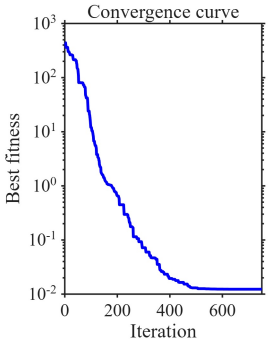

F11
Fitness
100
50
0
500
0
-500
x2
-500
0
500
x1


Search history
x2
600
400
200
0
-200
-400
-600
-500
0
500
x1


Average fitness
Fitness
1000
800
600
400
200
0
0
200
400
600
Iteration


First-agent trajectory
x1
500
0
-500
0
250
750
Iteration


Convergence curve
Best fitness
10^3
10^2
10^1
10^0
10^-1
10^-2
0
200
400
600
Iteration